Nick Kashani Motlagh

Machine Learning Researcher and Engineer | Ph.D. in Computer Science and Engineering
Columbus, OH | nmotlagh@gmail.com | +1 (614) 886-1412 | U.S. Citizen
nmotlagh.github.io | github.com/nmotlagh | linkedin.com/in/nicholas-kashani-motlagh
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Developed machine learning methods for computer vision and natural language processing. Four first-author peer-reviewed papers and a Springer **Best Paper Award**.

Education

Ohio State University — Columbus, OH
Ph.D., Computer Science and Engineering, GPA 3.70 — Aug 2026
Dissertation: *Answering Under Uncertainty: Abstention, Ambiguity, and Recoverability*
M.S., Computer Science and Engineering — Dec 2025
B.S. with Honors, Computer Science and Engineering, GPA 3.86 — May 2021

Experience

Computer Engineer II — May 2025 – Present
DCS Corporation, supporting the Air Force Research Laboratory — *Dayton, OH*

- Develop synthetic question-answer datasets from domain-specific documents and train QA models.
- Own the full pipeline from data generation through training and evaluation, using Slurm in high-performance computing environments.

Graduate Research Associate — Aug 2021 – Aug 2026
Computer Vision Lab, Ohio State University — *Columbus, OH*

- Developed per-class abstention methods without preset rejection costs or accuracy/coverage targets (ISVC 2022; MVA 2025): 97.4% selective accuracy at 79.7% coverage on ImageNet validation.
- Designed ImageCoMMuTE (WMT 2024), a metric for translation models facing ambiguity; compared confidence in fixed translations under matched and mismatched images across three English-to-French model families.
- Implemented paired-outcome evaluation pipelines to measure recoverability in retrieval-augmented QA; fine-tuned return-or-revise policies and evaluated 25,870 held-out questions (Return or Revise?, preprint). Work spanned OSU and DCS; completed at DCS.

Research Intern (graduate and undergraduate) — Summers 2020–2024
Air Force Research Laboratory — *Dayton, OH*

- Adapted and trained self-supervised multimodal models; implemented reject-option methods for translation and classifier training.
- Built an OpenStreetMap-guided pipeline for temporal satellite imagery collection, published at the ICCV 2021 workshop with public code.

Publications (first author)

- *Learning When to Say "I Don't Know"*. ISVC, 2022. Springer **Best Paper Award**.
- *Naturally Constrained Reject Option Classification*. Machine Vision and Applications, 2025.
- *Assessing the Role of Imagery in Multimodal Machine Translation*. WMT, 2024.
- *Return or Revise? Learning When Revision Helps Retrieval-Augmented QA*. arXiv:2609.30087, 2026 (preprint).
- *A Framework for Semi-Automatic Collection of Temporal Satellite Imagery for Analysis of Dynamic Regions*. ICCVW, 2021.

Teaching

Graduate Teaching Associate and Course Grader, Ohio State University — 2023–2026
Led Excel labs (Spring 2026); held office hours and graded for Machine Learning and Foundations of NLP (2023–2025).

Skills and Service

Language: Python **Tools used:** PyTorch, Hugging Face Transformers, Slurm
ML engineering: Fine-tuning, retrieval-augmented generation, distributed training, inference serving
Service: Reviewer for CVPR (2022, 2023), ICCV (2023), and ECCV (2022)